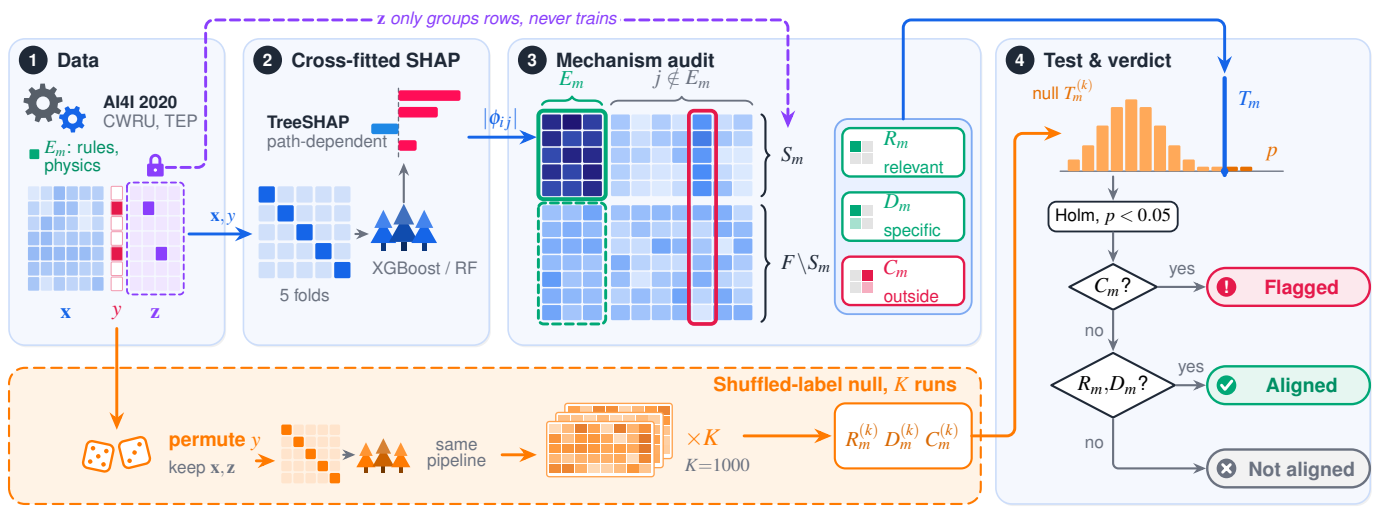

z only groups rows, never trains
1 Data
AI4I 2020
CWRU, TEP
E_m: rules, physics
x
y
z
x, y
2 Cross-fitted SHAP
TreeSHAP
path-dependent
XGBoost / RF
5 folds
$|\phi_{ij}|$
3 Mechanism audit
E_m
$j \notin E_m$
S_m
$F \setminus S_m$
R_m relevant
D_m specific
C_m outside
4 Test & verdict
null $T_m^{(k)}$
T_m
p
Holm, $p < 0.05$
C_m?
yes
Flagged
no
R_m, D_m?
yes
Aligned
no
Not aligned
Shuffled-label null, K runs
permute y
keep x, z
same pipeline
$\times K$
K=1000
$R_m^{(k)}$ $D_m^{(k)}$ $C_m^{(k)}$